Zhanhong He

Music and Audio Machine Learning Researcher

zhanh.he@research.uwa.edu.au | github.com/zhanh-he | linkedin.com/in/zhanhong-he

Profile

Music and audio ML researcher with experience in music-model post-training, generative modeling and perceptual evaluation. Research spans expressive musical dynamics, audio-symbolic conditioning and transferable supervision. Research internships at Huawei and Dolby complement public PyTorch research projects.

Research Experience

Research Intern | Huawei Hong Kong Research Centre | Started Jun 2026

- Applied DPO and GRPO in music-model post-training and contributed to rollout workflows.
- Developed music-oriented reward signals and contributed to human-preference data collection and analysis.

Research Intern | Dolby Laboratories, Sydney | May-Sep 2024

- Developed music generative models using U-Net, VAE and masked Transformer architectures.
- Designed and conducted professional listening tests to evaluate musical expression and model outputs.

Visiting Student Researcher, part time | Great Bay University | Started Mar 2026

- Explored pretrained music representations, including MERT, and agentic research workflows for music information retrieval.

Selected Research And Code

Piano dynamics and metrical structure

- Developed a multitask CRNN for dynamic markings, beats, downbeats and structural changes.
- Reimplemented a Bark-scale specific-loudness feature extractor in PyTorch.
- Code: github.com/zhanh-he/piano-dynamic-estimation

Cross-instrument MIDI velocity estimation

- Adapted piano-pretrained velocity estimation to other instruments using differentiable SoundFont proxies and audio-loudness supervision, reducing dependence on paired velocity labels.
- Project repository: github.com/zhanh-he/sfproxy-velocity-estimation

MIDI velocity generation and score-conditioned refinement

- Adapted an image-colorization U-Net to fill per-note MIDI velocity and designed listening tests with professional listeners.
- Developed a score-informed Transformer correction module for velocity estimates from piano transcription systems.
- Code: github.com/zhanh-he/midi-velocity-colorizer
- Code: github.com/zhanh-he/score-informed-amt

Education

PhD research in Electrical and Electronic Engineering | University of Western Australia | Feb 2022-present

Defense completed; graduation certificate expected November 2026. Formal award pending confirmation.

Research focus: musical dynamics, multitask learning and audio-symbolic music processing.

BSc Honours in Engineering Science | University of Western Australia | 2018-2021

First Class Honours | GPA 7/7 | WAM 87.7/100

Thesis: Improving Spectrogram-based Music Source Separation using Phase Information.

Selected Publication

He, Z., Meng, H., Huang, D. and Togneri, R. Joint Estimation of Piano Dynamics and Metrical Structure with a Multi-Task Multi-Scale Network. ICASSP 2026. DOI: 10.1109/ICASSP55912.2026.11460839.

Technical Skills

Python, PyTorch, TensorFlow; MATLAB. Music generation, music information retrieval, audio-symbolic modeling, preference-based post-training and perceptual evaluation. Ableton Live, Max/MSP and MuseScore. Basic C/C++.

Teaching And Collaboration

Tutor and Lab Facilitator | University of Western Australia | Feb 2023-Dec 2025

- Taught signal processing and engineering practice; co-supervised computer science honours projects.